Module 3, Transient Behavior and Initial Conditions

Transient Behavior & Initial Cond

Networks can be represented using integro-differential equation. The general soln of differential equation consists of complementary function & particular integral. Complementary function represents transient response. Particular soln represents steady state response.

* n^{th} order complementary fn in differential equation consists of 'n' no of constants. To evaluate these constants 'n' number of initial conditions are required.

Initial condn: Initial condn of the N/w or element are the condn of the N/w (voltage, or current) at the time of closing the switch at $t=0$.

$t=0$ is taken as reference.

* Immediately before switching operation these quantities are referred $v(0-)$, $i(0-)$, $q(0-)$.
* Immediately after switching operation these quantities are referred to as $v(0+)$, $i(0+)$, $q(0+)$.

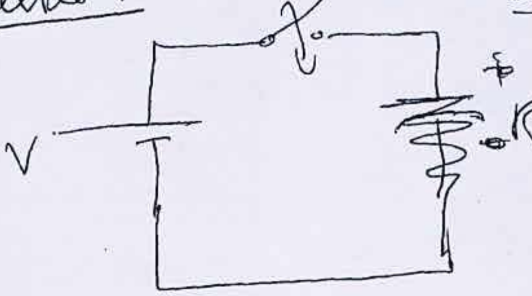
why do we need I.C's:

1. They are necessary to evaluate constants during soln of a differential equation.
2. The knowledge of the initial values of one or more derivatives of a response, are helpful in anticipating the form of response.
3. useful in understanding the behaviour of the element individually & in ...

Initial Conditions in elements: / July 2009 (8M) (2k)

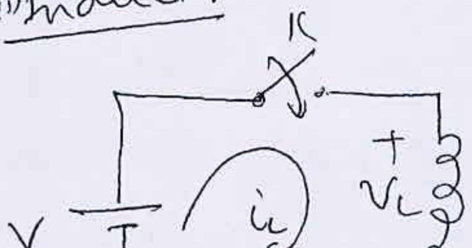
(i) Resistor   Explain transient behavior of R,

resistors cannot store energy



$\therefore i_R = \frac{V}{R}$

(ii) Inductor



Voltage across inductor is given by $V_L = L\frac{di_L}{dt}$

When D.C. is apply $\frac{di_L}{dt} = 0$.

$\therefore$ Under steady state Inductor acts as short ckt.

$\therefore$ voltage across it is '0'

$$i_L = \frac{1}{L}\int_{-\infty}^{t} v_L\,dt = \frac{1}{L}\int_{-\infty}^{0^-} v_L\,dt + \frac{1}{L}\int_{0^-}^{t} v_L\,dt$$

first term in the RHS indicates initial value of current through inductor before closing the switch. i.e. $i_L(0-)$

when switch is closed.

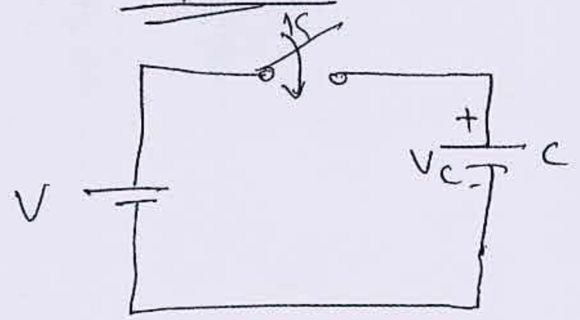
$$i_L(0+) = i_L(0-) + \frac{1}{L}\int_{0^-}^{0^+} V_L\,dt.$$

switching operation doesnot consume any time

$$\therefore \boxed{i_L(0+) = i_L(0-)}$$

i.e at $t=0+$, the inductor acts as open as the current doesnot change instantaneously. If initial current is present in the ~~circuit~~ inductor, immediately after closing the switch at $t=0+$ it acts as current source.

(iii) Capacitor



The current through the Capacitance is given

$$i_C = C \cdot \frac{dv_C}{dt}$$

If a D.C. voltage is app

$\frac{dv_C}{dt} = 0$, $i_C = 0$, Thus for D.C. quantities capacitance acts as opencircuit.

voltage across the capacitance is given by

$$v_C = \frac{1}{C}\int_{-\infty}^{t} i_C\,dt = \frac{1}{C}\int_{-\infty}^{0-} i_C\,dt + \frac{1}{C}\int_{0^-}^{t} i_C\,dt.$$

$$\frac{1}{C}\int_{-\infty}^{0-} i_C\,dt = V_C(0-) \text{ which is constant.}$$

when the Switch 'K' is closed at $t=0$.

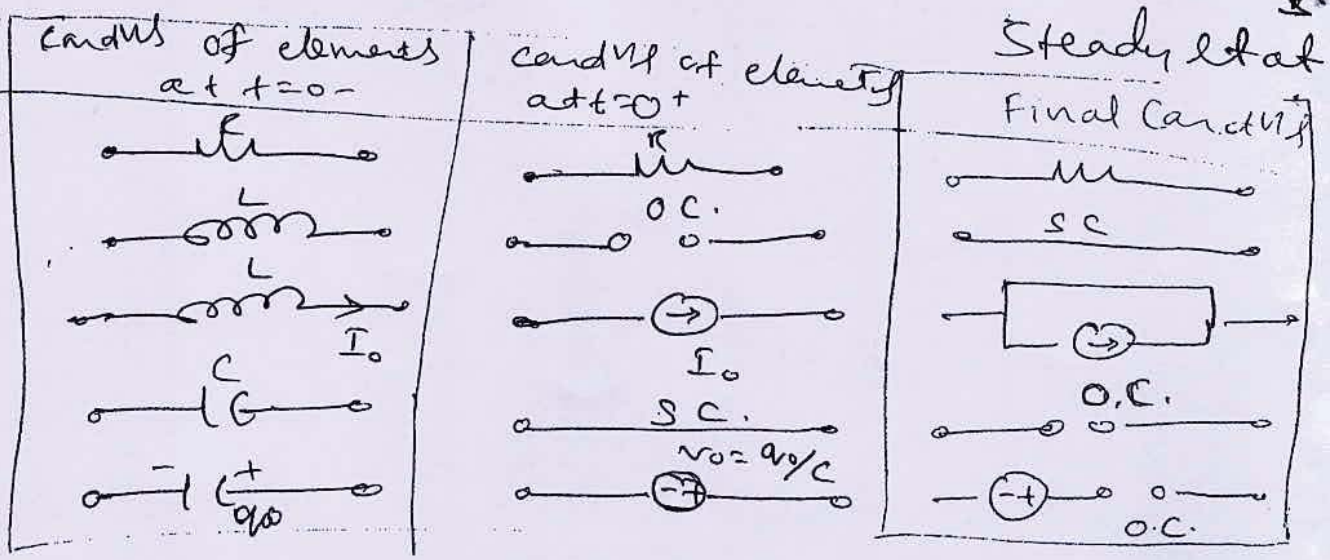
$$V_C(0+) = V_C(0-) + \frac{1}{C}\int_{0-}^{0+} i_C\,dt = V_C(0-) + 0$$

i.e. $\boxed{V_C(0+) = V_C(0-)}$

i.e. voltage across Capacitor cannot change insta-ntaneously. Hence Capacitor does not have any charges ~~before~~ $t=0-$, then at $t=0+$ also its voltage will be '0' ∴ Capacitor acts short ckt at $t=0+$ if it does not have intial charge at $t=0-$.

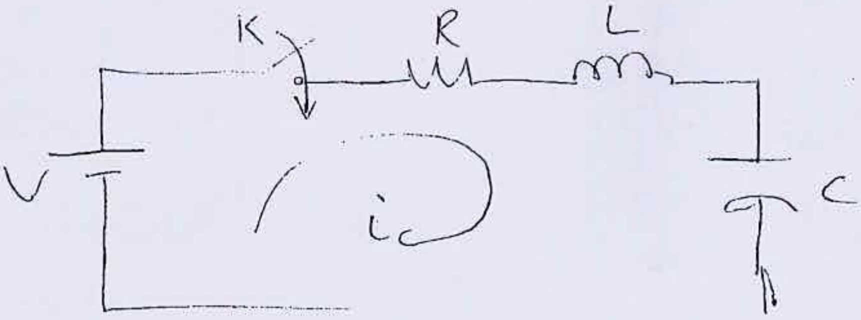
the Capacitor has initial voltage 'V_0'. then at $t=0+$ it actes as voltage source

Initial Conditions



Problems on Transient behaviour & Initial conditions (1)

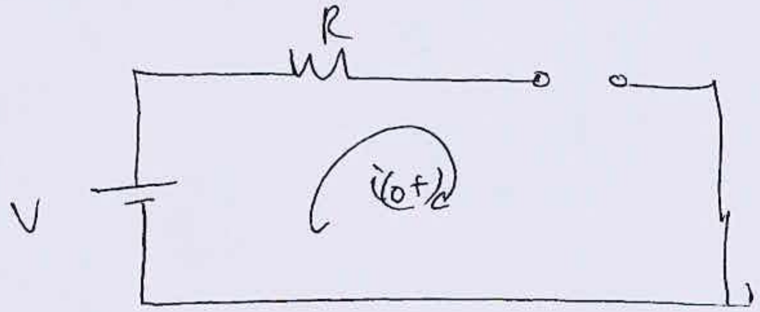
(1) In the circuit shown in fig $V = 10V$, $R = 10\Omega$, $L = 1$, $C = 10\ \mu F$ & $V_C(0+) = 0$, Find $i(0^+)$, $\frac{di}{dt}(0^+)$ & $\frac{d^2 i}{dt^2}(0^+)$



When 'K' is closed at $t=0$

$$V = Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt \quad — (1)$$

at $t = 0^+$, the network is as shown in fig.



Capacitor open
Inductor short

$$\boxed{i(0+) = 0} \quad (2)$$

Sub (2) in (1)

$$V = R\,i(0+) + L\cdot\frac{di}{dt}(0+) + \underbrace{\frac{1}{C}\int i(0+)\,dt}_{V_C(0+)}$$

given $V_C(0+) = 0$

$$\therefore V = R\,\cancel{i(0+)}^{\,0} + L\cdot\frac{di}{dt}(0+)$$

from (2)

$$\therefore \boxed{\frac{di}{dt}(0+) = \frac{V}{L} = \frac{10}{1} = 10\ A/sec}$$

Diff equ ①

$$R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i}{C} = 0$$

at $t = 0+$

$$R\frac{di}{dt}(0+) + L\frac{d^2i}{dt^2}(0+) + \cancel{\frac{i(0+)}{C}}^{0} = 0$$

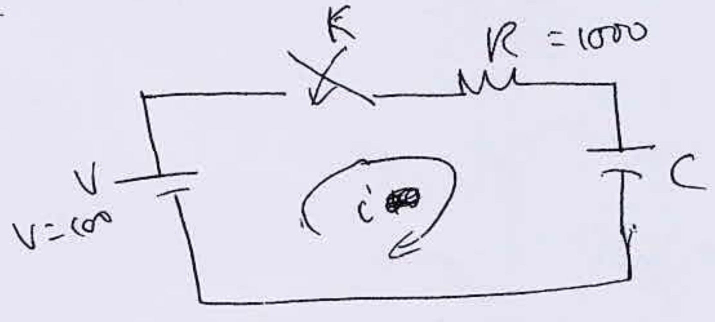
$$\therefore L\frac{d^2i}{dt^2}(0+) = -R\frac{di}{dt}(0+)$$

$$\therefore \frac{d^2i}{dt^2}(0+) = -\frac{R}{L}\frac{di}{dt}(0+)$$

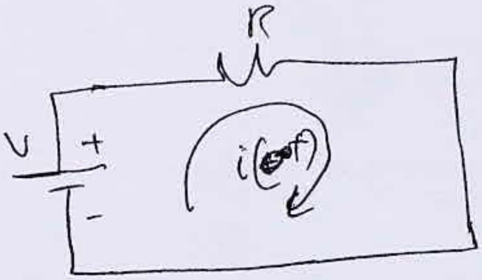
$$= -\frac{10}{1}\,10$$

$$\boxed{\frac{d^2i}{dt^2}(0^+) = -100\ A/sec^2}$$

② In the network shown in fig. the switch K is closed at $t=0$, with the capacitor uncharged. Find the values of i, $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$, at $t = 0+$, for element values as follows, $V = 100V$, $R = 1000\Omega$, $C = 1\mu F$



at $t = 0+$ eqt circuit is as follows



$$i(0+) = \frac{100}{1000}$$

$$\boxed{i(0^+) = 0.1\ A}$$ — ①

when K is closed at t=0.

$$V = Ri + \frac{1}{C}\int i\,dt$$

on Differentiation $\quad 0 = R\frac{di}{dt} + \frac{i}{C}$ — (2)

at $t=0+$

$$0 = R\frac{di}{dt}(0+) + \frac{i}{C}(0+)$$

$$\frac{di}{dt}(0+) = -\frac{i}{C}(0+)\cdot\frac{1}{R} = -\frac{0.1}{1\times10^{-6}}\cdot\frac{1}{1000}$$

$\boxed{\frac{di}{dt}(0+) = \cancel{-0.1\,mA/sec}}$ $\boxed{-100\ A/sec}$

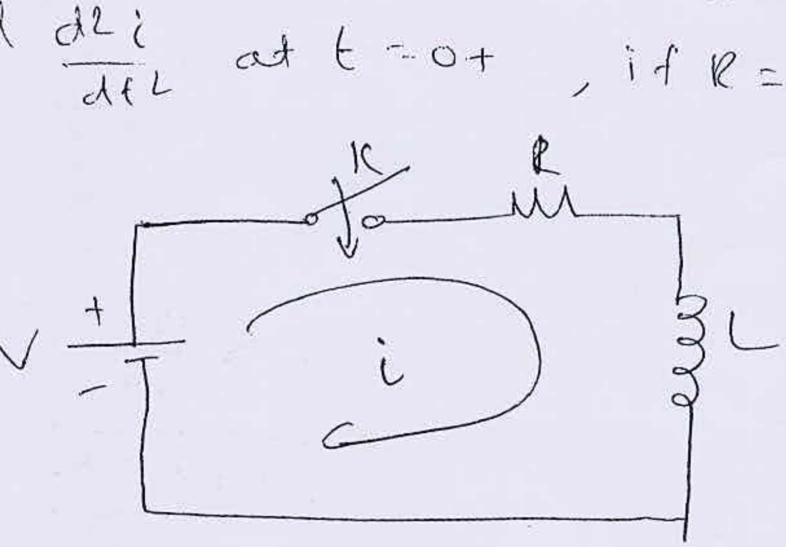
Diff (2) $\quad R\frac{d^2i}{dt^2} + \frac{1}{C}\frac{di(0+)}{dt} = 0$

at $t=0+$ $\quad R\frac{d^2i}{dt^2}(0+) = -\frac{1}{C}\frac{d}{dt}(i(0+))$

$$\frac{d^2i}{dt^2}(0+) = -\frac{1}{RC}\frac{di(0+)}{dt} = \cancel{-}\frac{1}{1000\times10^{-6}}\cdot(-100)$$

$$\boxed{\frac{d^2i}{dt^2}(0+) = 0.1\times10^{6}\ A/m2}$$

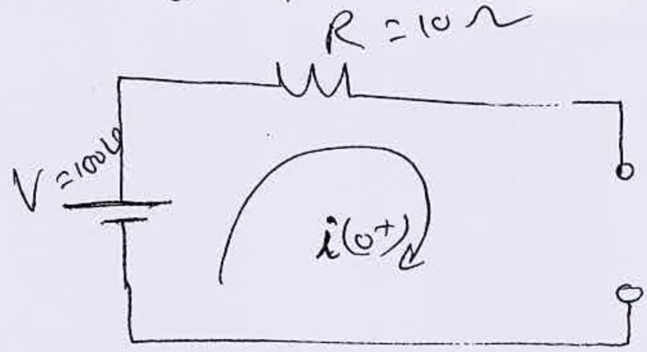
3) In the network shown in fig., K is closed at t with zero current in the inductor. Find i, $\frac{di}{dt}$, and $\frac{d^2i}{dt^2}$ at $t=0+$, if $R=10\Omega$, $L=1H$ and $V=100V$.



when K is closed at $t=0$

$$V = Ri + L\frac{di}{dt} \quad \text{—(1)}$$

at $t=0+$



eq^t ckt at $t=0^+$

is as shown.

$\therefore \boxed{i(0+) = 0}$

from ① ~~at t=0~~ ~~differentiating~~ ~~w.r.t.~~ $\cancel{0 = R\frac{di}{dt}}$

$L\frac{di}{dt} = V - Ri$ — ② $\therefore \frac{di}{dt} = \frac{V - Ri}{L}$

at $t=0+$ $\frac{di(0+)}{dt} = \frac{V - Ri(0+)}{L}$

$\frac{di}{dt}(0+) = \frac{100 - 10(0)}{1} = \boxed{100\,A/sec}$

Diff ② $L\frac{d^2i}{dt^2} = -R\frac{di}{dt}$

at $t=0+$ $\frac{d^2i(0+)}{dt^2} = -\frac{R}{L}\frac{di(0+)}{dt}$

Aug2004 (10M) III^r

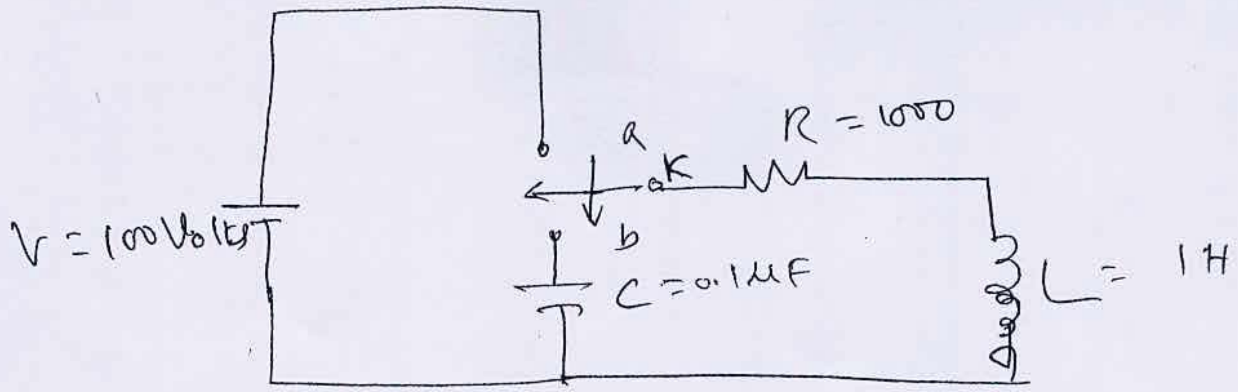
Jan2009 III^r prob.

June 2010 (10M)

$$\boxed{\frac{d^2i}{dt^2}(0+) = -\frac{10}{1}\,100 = -1000\,A/sec^2}$$

(4) In the network shown in fig. K is changed from position a to b at $t=0$. Solve for i, $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0+$, if $R = 1000\Omega$, $L = 1H$, $C = 0.1$ and $V = 1000V$. Assume that the capacitor is initially uncharged.

(c)



When the switch is 'K' is at 'a' inductor acts a short ckt (network is under steady state)

$$i(0-) = \frac{V}{R} = \frac{100}{1000} = \boxed{0.1\ A}$$

$\therefore$ $i(0+) = 0.1 A$

i.e. the inductor has an initial current of 0.1

when 'K' is changed from a to b, $i(0+) = 0.1 A$

when 'K' is at 'b'

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = 0 \quad -(3)$$

at $t = 0+$, $R\,i(0+) + L\cdot\frac{d\,i(0+)}{dt} + \frac{1}{C}\int i(0+)\,dt = 0.$

it is given that $\frac{1}{C}\int i(0+)\,dt = V_C(0+) = 0.$ (Capacitor is initially uncharged)

$$L\frac{di}{dt}(0+) = -R\,i(0+)$$

$$\frac{di}{dt}(0+) = -\frac{R}{L}\,i(0+) = -\frac{1000}{1}(0.1) = -100 A$$

Diff (3). $R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i}{C} = 0$

at $t = 0+$

$$R\frac{d}{dt}i(0+) + L\frac{d^2}{dt^2}i(0+) + \frac{i(0+)}{C} = 0$$

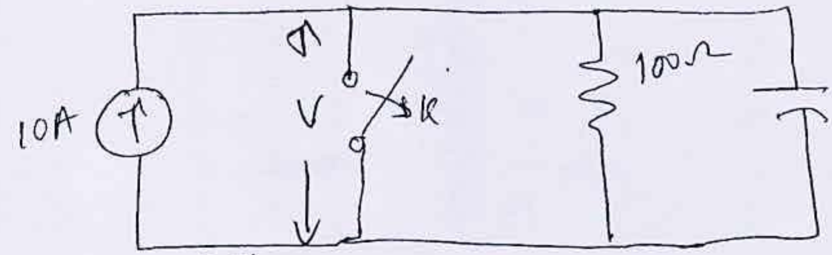
$$R.\ 1000(-100) + L\frac{d^2i(0+)}{dt^2} + \frac{0.1}{0.1\times10^{-6}} = -10^6$$

at

$$L\frac{d^2i(0+)}{dt^2} = -10^6 + 10^5 = -9\times10^5$$

$$\therefore \frac{d^2i}{dt^2} = -\frac{11\times10^5}{} = -9\times10^5\ A/(\text{sec})$$

(5) In the circuit shown in the figure the switch (K) 'L' is opened at $t=0$, find the values of V, $\frac{dv}{dt}$ & $\frac{d^2V}{dt^2}$ at $t=0+$



When the switch 'K' is closed all the current flow through the S.C. The capacitor is not charged as it acts as an O.C.

$$V_C(0-) = 0 = V_C(0+)$$

When 'K' is opened.

$$-10 + \frac{V}{100} + C\frac{dV}{dt} = 0 \quad \therefore \frac{V}{100} + C\frac{dV}{dt} = 10$$

$$C\frac{dv}{dt} = 10 - \frac{V}{100}$$

$$\therefore \frac{dV}{dt} = \frac{10}{C} - \frac{V}{100C} \quad — (1)$$

at $t=0+$

$$\frac{dV}{dt}(0+) = \frac{10}{10^{-6}} - \frac{V(0+)}{100\times 10^{-6}}$$

$$\boxed{\frac{dV}{dt}(0+) = 10^{+7}\ V/sec}$$

Diff (1) w.r to 't'

$$\frac{d^2v}{dt^2} = 0 - \frac{1}{100C}$$

$$\frac{d^2V}{dt^2} = -\frac{1}{100C}\frac{dV}{dt}$$

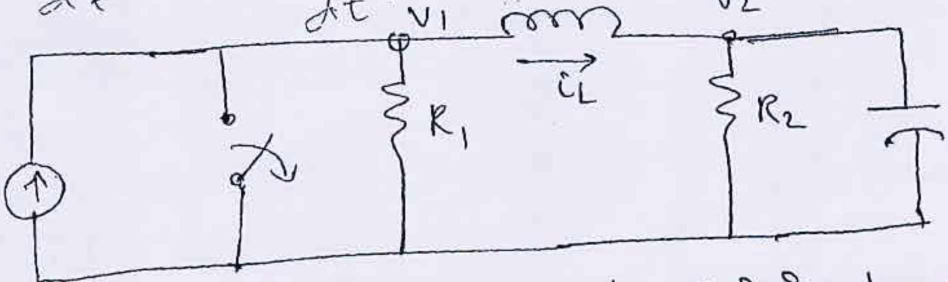
at $t=0+$

$$\frac{d^2V(0+)}{dt^2} = \frac{-1}{100\times 10^{-6}}\frac{dV}{dt}$$

$$= \frac{-1}{100\times 10^{-6}}\times 10^{7} = -10^{+11}\ V/.$$

(6) ~~In the network shown in fig 'K' is changed (from the position a to b at t=0. Solve for 'i';~~ ~~$\frac{di}{dt}$~~ ~~July~~ May/ June 2010 (10M)

(6) In the network shown in fig. ~~K is change~~ ~~from~~ Po has two ~~two~~ independent node pairs. If the switch 'K' is opened at t=0, find the following quantities at $t=0^+$, (i) V_1 (ii) V_2 (iii) $\frac{dV_1}{dt}$ (iv) $\frac{dV_2}{dt}$ (v) $\frac{di_L}{dt}$



When the 'K' is closed, all the current flows through S.C. & hence $i_L(0-)=0$, $V_C(0-)=0$

$$\therefore i_L(0+)=0, \quad \therefore V_C(0+)=0$$

$$\therefore \boxed{V_2(0+)=0}$$

When K is opened at t=0.

For node V_1, $\quad i(t) = \frac{V_1}{R_1} + i_L \quad -(1)$

at t=0+, $\quad i(0+) = \frac{V_1(0+)}{R_1} + 0 \quad \therefore \boxed{V_1(0+) = i(0+)R_1}$

For Node (2). $\quad -i_L + \frac{V_2}{R_2} + C\frac{dV_2}{dt} = 0$

at t=0+ $\quad , -i_L(0+) + \frac{V_2(0+)}{R_2} + C.\frac{dV_2}{dt}(0+) = 0$

$$0 + 0 + C\frac{dV_2}{dt}(0+) = 0$$

$$\therefore \boxed{\frac{dV_2}{dt}(0+) = 0}$$

Diff. eqn ① $\frac{di}{dt} = \frac{1}{R_1}\frac{dv_1}{dt} + \frac{di_L}{dt}$.

$$\frac{di}{dt}(0+) = \frac{1}{R_1}\frac{dv_1(0+)}{dt} + \frac{di_L(0+)}{dt}$$

$$\frac{1}{R_1}\frac{dv_1(0+)}{dt} = \frac{di}{dt}(0+) - \frac{di_L}{dt}(0+) \quad \text{—} ②$$

~~$\frac{1}{R_1}\frac{dv_1(0+)}{dt} =$~~ But $i_L = \frac{1}{L}\int (v_1 - v_2)\,dt$

Diff $\frac{di_L}{dt} = \frac{1}{L}(v_1 - v_2)$

at $t = 0+$

$$\frac{di_L}{dt}(0+) = \frac{1}{L}\left[v_1(0+) - v_2(0+)\right]$$

$$\boxed{\frac{di_L}{dt}(0+) = \frac{1}{L}\left[i(0+)R_1 - 0\right]}$$

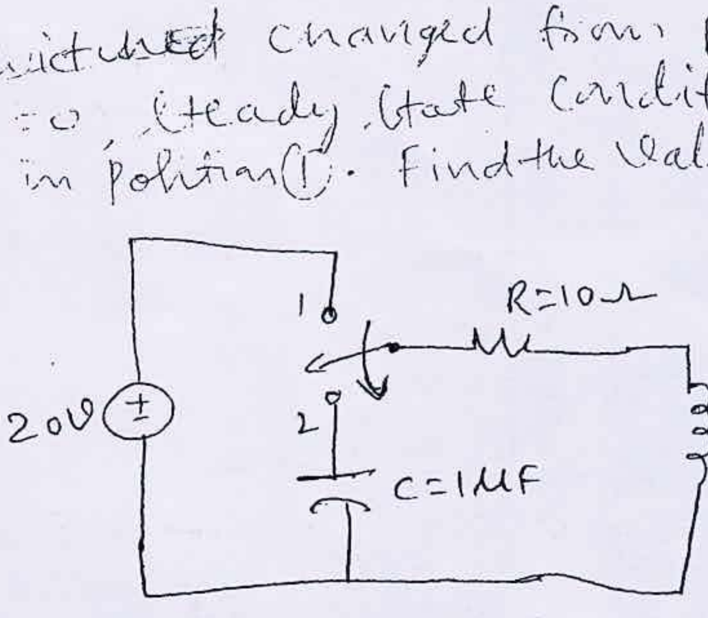
Sub in ②

$$\therefore \frac{1}{R_1}\frac{dv_1(0+)}{dt} = \frac{di}{dt}(0+) - \frac{1}{L}\left[i(0+)R_1\right]$$

$$\therefore \boxed{\frac{dv_1}{dt}(0+) = R_1\left[\frac{di}{dt}(0+) + \frac{1}{L}R_1\, i(0+)\right]}$$

⑦ Jan 2007 (10M)

July 2009 | For the circuit shown in fig, the switch K is ~~switched~~ changed from position ① to position ② at $t=0$, steady state condition having been reached in position ①. Find the values of i, $\frac{di}{dt}$, & $\frac{d^2i}{dt^2}$ at $t=0$



When the switch is at ①, the ckt is under steady state conditions., L acts as a S.C. & C acts as O.C.

$$i(0-) = \frac{V}{R} = \frac{100}{10} = 10\,A = i(0+)$$

$$V_c(0-) = 0 = V_c(0+)$$

when the switch is changed from ① to ②

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = 0 \quad -①$$

at $t=0+$

$$R\,i(0+) + L\frac{di}{dt}(0+) + \frac{1}{C}\int i(0+)\,dt = 0$$

$$(10)(100) + 1\,\frac{di}{dt}(0+) + V_c(0+) = 0$$

$$\boxed{\frac{di}{dt}(0+) = -1000\,A/sec}$$

Diff ①

$$R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i}{C} = 0$$

at $t=0+$

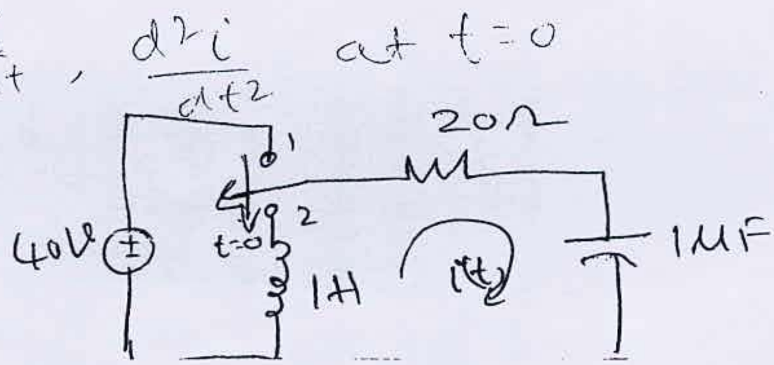
$$R\frac{di}{dt}(0+) + L\frac{d^2i}{dt^2}(0+) + \frac{i(0+)}{C} = 0$$

$$10\,(-1000) + 1\cdot\frac{d^2i}{dt^2}(0+) + \frac{10}{10^6} = 0$$

$$\boxed{\frac{d^2i}{dt^2}(0+) = 10^7 + 10^4 = 10^4(1010)\ A/sec^2}$$ verify

June 2012 (or)

3) Jan 2009 For the network shown in fig. the switch is moved from position 1 to position ② at $t=0$, steady state has been reached before switching. Calculate i, $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t=0$



When the switch is on 1, 'L' acts as S.C. & C acts as O.C. as the circuit is under steady state condn.

at $t=(0-)$, $\boxed{i(0-) = \frac{20}{10} = 2A = i(0+)}$

$V_C(0-) = 0 = V_C(0+)$

When 'K' is changed from (1) to (2),

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = 0 \quad -①$$

at $t = 0+$

$$Ri(0+) + L\frac{di}{dt}(0+) + \underbrace{\frac{1}{C}\int i(0+)\,dt}_{V_C(0+)=0} = 0$$

$$\therefore L\frac{di}{dt}(0+) = -R\,i(0+) = -(10)(2) = -20$$

$$\therefore \boxed{\frac{di}{dt}(0+) = \frac{-20}{L} = \frac{-20}{1} = -20\,A/sec.}$$

Diff ①

$$R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{1}{C}i = 0$$

$$L\frac{d^2i}{dt^2} = -\frac{i}{C} - R\frac{di}{dt}$$

at $t=0+$

$$L\frac{d^2i}{dt}(0+) = -\frac{i(0+)}{C} - R\frac{di}{dt}(0+)$$

$$\frac{d^2i}{dt^2}(0+) = -\frac{i(0+)}{LC} - \frac{R}{L}\frac{di}{dt}(0+)$$

$$= -\frac{2}{(1)(1\times10^{-6})} - \frac{10}{1}(-20)$$

$$= -2\times10^6 + 200$$

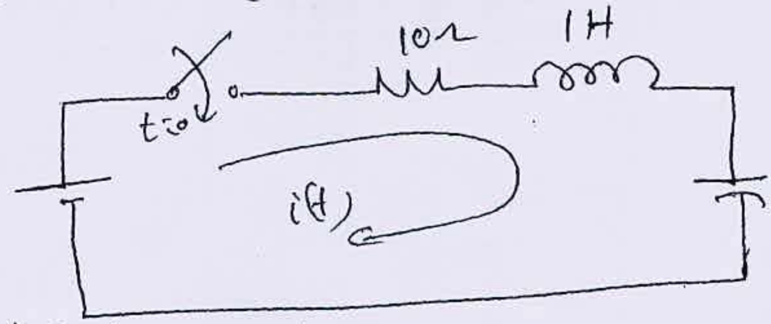
$$= -2000000 + 200$$

$$\boxed{\frac{d^2i}{dt^2}(0+) = -19,998,00\ A/s^2}$$

(9) For the network shown in f find out $i(0^+)$, $\frac{di}{dt}(0^+)$ & $\frac{d^2i}{dt^2}(0^+)$, take $V_c(0)=0$, if K is closed at $t=0$

July 2008 (7M)

Jan/Feb 2004 (6M)



cond^n at $t=0^+$

when 'K' is closed at $t=0$, L acts as 'O.C' & C acts as short circuit. $\therefore t=0^+$, $\boxed{i(0^+)=0}$ & $V_c(0^+)=0$

when switch closed at $t=0$, the general eq^n for the circuit is

$$R\,i(t)+L\frac{di}{dt}+\frac{1}{C}\int i\,dt = 10 \quad -(1)$$

at $t=0^+$

$$R\,\cancel{i(0^+)}^{\,0}+L\frac{di}{dt}(0^+)+\underbrace{\frac{1}{C}\int i(0^+)dt}_{V_c(0^+)=0}=10$$

$$\therefore \frac{di}{dt}(0^+)=\frac{10}{L}=\frac{10}{1}=\boxed{10\ A/sec}$$

Diff (1)

$$R\frac{di(t)}{dt}+L\frac{d^2i}{dt^2}+\frac{1}{C}i(t)=0.$$

at $t=0^+$.

$$L\frac{d^2i}{dt^2}(0^+)=-\frac{1}{C}\cancel{i(0^+)}^{\,0}-R\frac{di}{dt}(0^+)$$

$$\boxed{\frac{d^2i(0^+)}{dt^2}=-\frac{10}{1}10=-100\ A/sec^2}$$

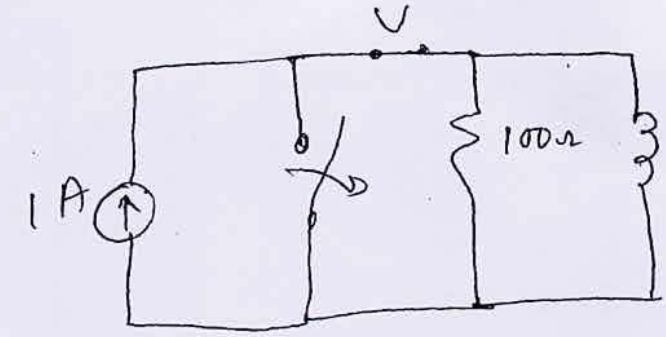
Jan 2006 (6M)

(10) A parallel R-L circuit is energized by a current source of 1A. The switch across the source is opened at $t=0$, Solve for V, DV, D^2V all at $t=0^+$, if $R=100\Omega$ & $L=1H$.

$V(0^+)$

$DV(0^+)$

$D^2V(0^+)$



When the switch is closed, all the current flows through short circuit. (12

Hence $i_L(0-)=0$, $V(0-)=0$

when switch is opened at $t=0$, L acts as an O.C. all the current flows through resistor
($t=0+$ cond'n)

$$\therefore \boxed{V(0+) = (1)\,100 = 100 \text{ Volts}}$$

The general eqn at $t=0$ is given by

$$i(t) = \frac{V}{100} + \frac{1}{L}\int v\,dt \quad \text{— ①}$$

at $(t=0+)$, $i(0+) = \frac{V(0+)}{100} + \frac{1}{1}\int V(0+)\,dt$

diff. $\frac{di}{dt}(0+) =$ $\quad 0 = \frac{1}{100}\frac{dV(t)}{dt} + \frac{1}{L}V(t) \quad$ —②

Diff ① at $t=0+$

$$\frac{1}{100}\frac{dV(0+)}{dt} = -\frac{1}{L}V(0+)$$

$$\therefore \frac{dV}{dt}(0+) = -\frac{100}{1}\times 100 = -10{,}000 \text{ V/sec.}$$

Diff ②.

$$\frac{1}{100}\frac{d^2V(t)}{dt^2} + \frac{1}{L}\frac{dV(t)}{dt} = 0$$

$$\frac{1}{100}\frac{d^2V(t)}{dt^2} = -\frac{1}{L}\frac{dV(t)}{dt}$$

at $t=0+$

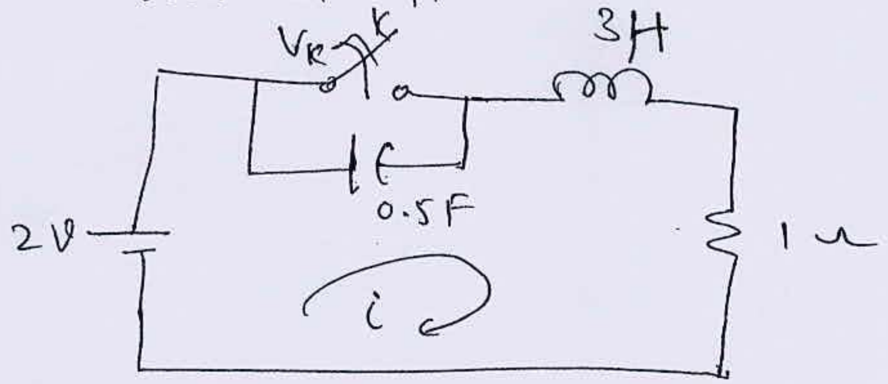
$$\frac{d^2V}{dt^2}(0+) = -\frac{100}{1}\frac{d}{dt}V(0+)$$

$$= -100\,(-10{,}000)$$

$$\boxed{\frac{d^2V}{dt^2}(0+) = 10^6 \text{ V/Sec}^2}$$

(11) Jan/Feb 2005 (TOM) Jan 2015

$R = 1\Omega$, $L = 1H$, & $C = \frac{1}{2}F$ are in series with a switch across 'C'. 2 volts is applied to circuit. At $t = 0-$, the switch is in closed. At $t = 0$ the switch is opened. Find at $t = 0+$, the voltage across the switch, its first and second order derivatives.



at $t = 0-$ There is a short circuit across capacitor

$\therefore V_C(0-) = 0 = \boxed{V_C(0+)}$.

Under steady state cond^n Inductance acts as short circuit.

$\therefore i(0-) = \frac{2}{1} = \underline{2A} = i(0+)$

when the switch is opened $V_K = V_C$

i.e. $V_K(0-) = \boxed{V_K(0+) = 0}$.

$V_C = \frac{1}{C}\int i\,dt = V_K$ —①

Diff.

$\frac{dV_K}{dt} = \frac{i}{C}$ —② $\therefore$ at $t = 0+$

$\frac{dV_K}{dt}(0+) = \frac{1}{C}\, i(0+)$

$\therefore \boxed{\frac{dV_K}{dt}(0+) = \frac{2}{0.5} = 4\ V/sec}$

general eqn is

$V = \frac{1}{C}\int i\,dt + L\frac{di}{dt} + Ri$

at $t=0+$

$$2 = \underbrace{\frac{1}{C}\int i(0+)\,dt}_{V_K(0+)} + L\frac{di}{dt}(0+) + Ri(0+) \qquad (1)$$

$$\cancel{2} = 0 + 3\cdot\frac{di}{dt}(0+) + 1\,(\cancel{2}) \qquad \therefore \boxed{\frac{di(0+)}{dt} = 0}$$

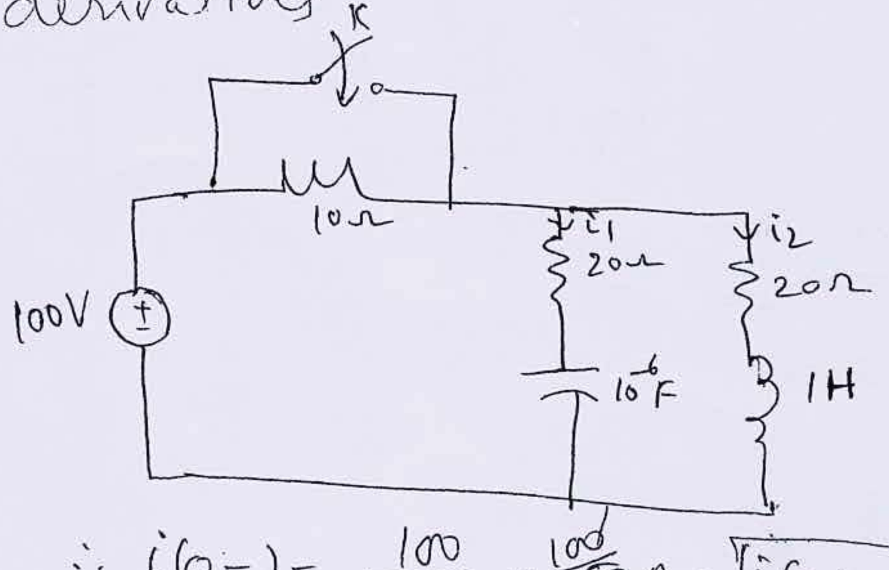
Diff (2) $\frac{d^2V_K}{dt^2} = \frac{1}{C}\frac{di}{dt}$

at $t=0+$ $\therefore \frac{dV_K(0+)}{dt} = \frac{1}{C}\,i(0+) = \frac{1}{0.5}\cdot 2 =$

at $t=0+$

Jan 2019 (10M) $\boxed{\frac{d^2V_K}{dt^2}(0+) = \frac{1}{C}\,\frac{di}{dt}(0+) = \frac{1}{0.5}\cdot(0) = 0}$

(12) Jan/Feb 2006 (8M) A series RLC circuit with $R = 20\Omega$ and $C = 1\mu F$ is shunted by an inductor of resistance 20Ω and inductance 1H. This is supplied by a D.C. source of 100V through a series resistance of 10Ω. There is a switch across 10Ω which is closed at $t=0$. Solve for the currents in L & C and their first derivatives



When 'K' is open ckt is under steady state case Inductor acts as short ckt & capacitor acts as open ckt

$$\therefore i_L(0-) = \frac{100}{20+10} = \frac{100}{30}\,A = \boxed{i_L(0+) = \frac{10}{3}\,A = i_2(0+)}$$

$i_C(0-) = 0, = i_1(0-) = i_1(0+)$

$V_C(0-) = 100 - (10)\left(\frac{10}{3}\right) = \frac{200}{3}$ Volts $V_C(0+)$ (15)

when 'K' is closed at $t=0$, the general equations are

$100 = 20 i_1 + \frac{1}{10^{-6}} \int i_1 \, dt$

at $t=0+$

$\cancel{100 = 20 i_1(0+) + V_C(0+)}$

$\therefore \cancel{i_1(0+) = \frac{100 - 10\left(\frac{10}{3}\right)}{20}} = \cancel{\frac{200/3}{20}} = \cancel{\frac{10}{3}}$

$20 i_1 + 10^6 \int i_1 \, dt = 100$ — ①

$\frac{di_2}{dt} + 20 i_2 = 100$ — ②

$\boxed{i_1(0+) = \frac{100 - \overbrace{10^6 \int i_1 \, dt}^{V_C(0+)}}{20} = \frac{100 - \frac{200}{3}}{20} = \frac{100/3}{20} = \frac{5}{3} A}$

$i_1(0+) = \frac{5}{3} A$ & $i_2(0+) = \frac{10}{3} A$ (known from ⓐ)

Diff ① $20 \frac{di_1}{dt} + 10^6 i_1 = 0$

at $t=0+$ $\frac{di_1(0+)}{dt} = \frac{-10^6 i_1(0+)}{20} = -\frac{10^5}{2}\cdot\frac{5}{3} = -\frac{5}{6} \times 10^5 A/sec.$

$$\boxed{\frac{di_1}{dt}(0+) = -83{,}333 \ A/sec}$$

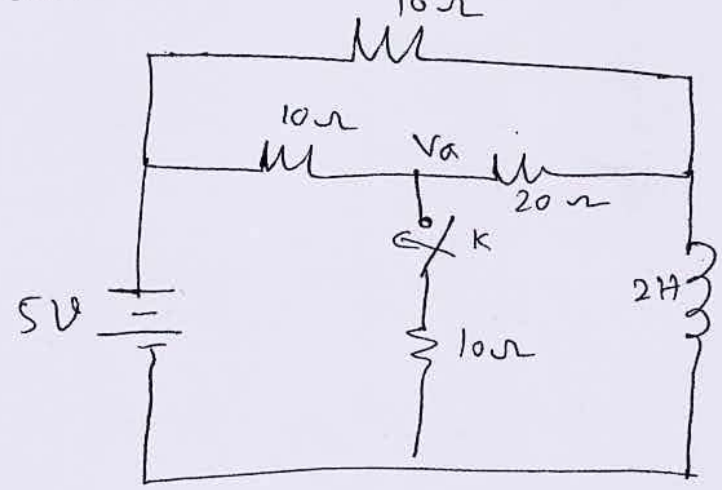
at $t=0+$

$\frac{di_2(0+)}{dt} = 100 - 20 i_2(0+) = 100 - 20\left(\frac{10}{3}\right)$

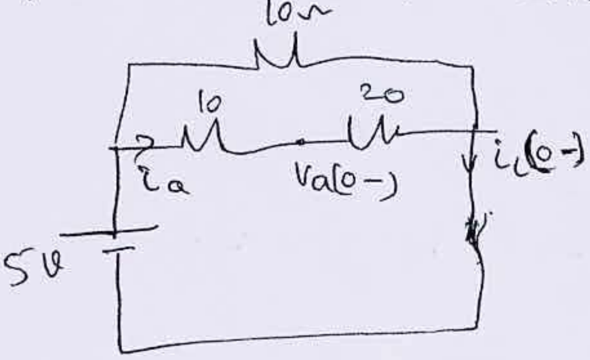
$$\boxed{\frac{di_2}{dt}(0+) = 33.33 \ A/sec}$$

~~[illegible]~~ July 2006 (CM)

(13) In the network shown in fig steady state is reached with the switch K open. At $t=0$, the switch is closed. For the element values given determine the values $V_a(0-)$ & $V_a(0+)$



when switch is open ckt is at steady state condn. inductor acts as short ckt



$\cancel{i_L(0-) = \frac{8}{630} + \frac{6}{10_2} = \frac{1}{6} + \frac{1}{2}}$

$\cancel{= \frac{1+3}{6}}$

$\cancel{\frac{4}{6} = \frac{2}{3}}$

$\cancel{i_L(0-) = 0.671}$

$\cancel{V_a(0-) = 0.67 \times 20}$

$\cancel{i_a = \frac{(0.67)10}{30} = 0.2233}$ | i_L

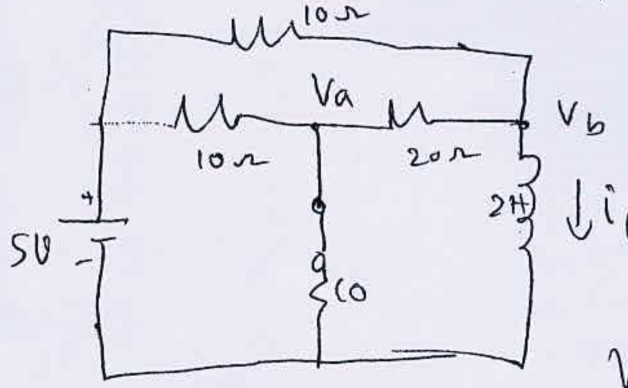
$\cancel{V_a(0-) = (0.2233)(20)}$

$$i_L(0-) = \frac{5}{\frac{10\times 30}{40}} = \frac{200}{10\times 30} = 0.67\ A = i_L(0+)$$

$$i_a(0-) = \frac{(0.67)\,10}{40} = 0.1675 \text{ Amps}$$

$$\therefore V_a(0-) = i_a(0-)(20) = 0.1675 \times 20 = \underline{3.35} \text{ volts}$$

When the switch K is closed at $t=0$



For node 'a' at $t=0+$

$$\frac{V_a(0+)-5}{10}+\frac{V_a}{10}+\frac{V_a(0+)-V_b}{20}=0$$

$$V_a(0+)\left[\frac{1}{10}+\frac{1}{10}+\frac{1}{20}\right]-\frac{V_b}{20}=\frac{5}{10}$$

× by 20 $\quad V_a(0+)\left[2+2+1\right]-V_b(0+)=10$

$$5V_a(0+)-V_b(0+)=10 \quad -(1)$$

For Node (b)

$$\frac{V_b(0+)-V_a(0+)}{20}+\frac{V_b-5}{10}+i_L(0+)=0$$

~~$V_b(0+)\left[\frac{1}{20}\right.$~~

$$-\frac{1}{20}V_a(0+)+\left[\frac{1}{20}+\frac{1}{10}\right]V_b(0+)=-i_L(0+)+\frac{5}{10}$$

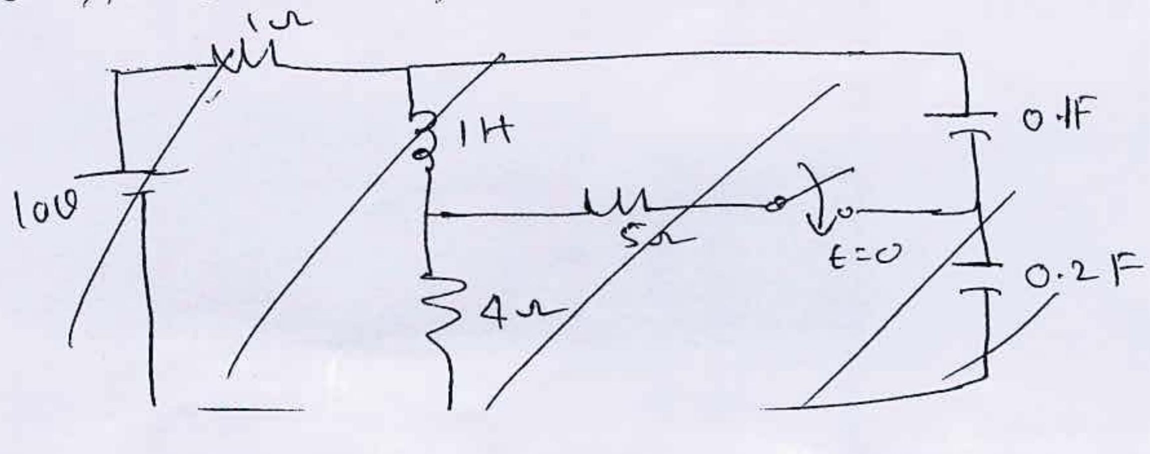
× by 20

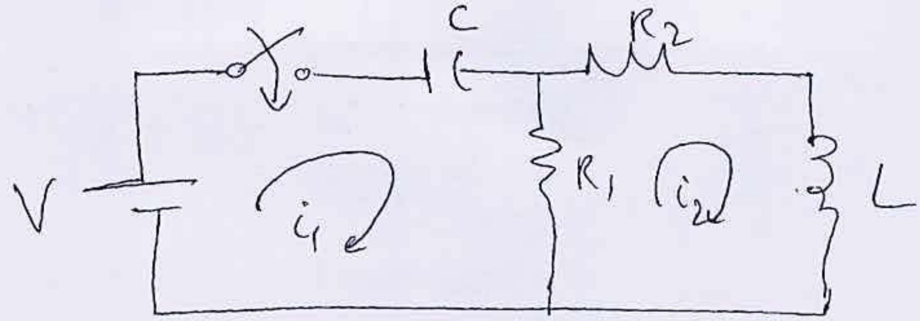
$$-V_a(0+)+\left[1+2\right]V_b(0+)=-20\,i_L(0+)+10$$

$$-V_a(0+)+3V_b(0+)=-20(0.67)+10=-3.4$$

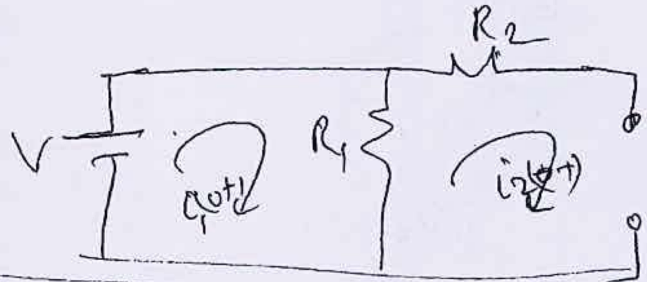
$$\boxed{V_a(0+)=1.9\text{ volts}}$$

14) Jan 2008 (10M) For the network shown in fig, find i_1, i_2, $\frac{di_1}{dt}$, $\frac{di_2}{dt}$, $\frac{d^2i_2}{dt^2}$ at $t=0+$. Circuit was in steady state before the closure of the switch. Assume all initial conditions zero.





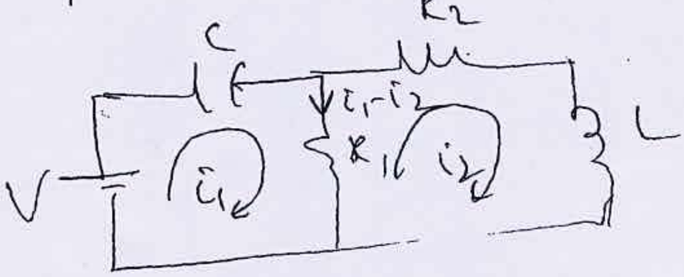
Network at $t = 0^+$, is as shown in fig.



$$\boxed{i_1(0^+) = \frac{V}{R_1}, \quad i_2(0^+) = 0}$$

writing loop equations for loop ① & ②

~~$R_1 i_1 + V = 0$~~



$$V = \frac{1}{C}\int i_1\,dt + R_1(i_1 - i_2) \quad — ①$$

$$-R_2 i_2 - L\frac{di_2}{dt} + R_1(i_1 - i_2) = 0 \quad — ②$$

Substituting I.C. in eqn ②

$$R_2 i_2(0^+) + L\frac{di_2}{dt}(0^+) = R_1\left[i_1(0^+) - i_2(0^+)\right]$$

$$0 + L\frac{di_2}{dt}(0^+) = R_1\left[\frac{V}{R_1}\right]$$

$$\therefore \boxed{\frac{di_2}{dt}(0^+) = + \frac{V}{L}}$$

Diff eqn ①

at $t=0+$,

$$\frac{i_1}{C} + R_1\left(\frac{di_1}{dt} - \frac{di_2}{dt}\right) = 0 \quad \text{—③}$$

$$\frac{i_1 0+}{C} + R_1 \frac{di_1}{dt}(0+) - R_1 \frac{di_2}{dt}(0+)$$

$$\therefore R_1 \frac{di_1(0+)}{dt} = R_1 \frac{di_2}{dt}(0+) - \frac{i_1(0+)}{C}$$

$$= R_1 \frac{V}{L} - \frac{V}{R_1 C}$$

$$\therefore \boxed{\frac{di_1}{dt}(0+) = \frac{V}{L} - \frac{V}{R_1^2 C}}$$

Diff eqn ②

$$-R_2\frac{di_2}{dt} - L\frac{d^2 i_2}{dt} + R_1\left[\frac{di_1}{dt} - \frac{di_2}{dt}\right] = 0$$

$$L\frac{d^2 i_2}{dt} = R_2\frac{di_2}{dt} + R_1\left(\frac{di_1}{dt} - \frac{di_2}{dt}\right)$$

at $t=0+$

$$L \cdot \frac{d^2 i_2}{dt^2} = R_2\frac{di_2}{dt}(0+) + R_1\frac{di_1}{dt}(0+) - R_1\frac{di_2}{dt}(0+)$$

$$= R_2\frac{V}{L} + R_1\left[\cancel{\frac{V}{L}} - \frac{V}{R_1^2 C}\right] - \cancel{R_1 \frac{V}{L}}$$

$$L\frac{d^2 i_2(0+)}{dt^2} = \frac{R_2 V}{L} - \frac{V}{R_1 C}$$

$$\boxed{\frac{d^2 i_2(0+)}{dt^2} = \frac{R_2 V}{L^2} - \frac{V}{R_1 L C}}$$

Differentiating eqn ③

$$\frac{1}{C}\frac{di_1}{dt} + R_1\left[\frac{d^2 i_1}{dt^2} - \frac{d^2 i_2}{dt^2}\right] = 0.$$

$$R_1\frac{d^2 i_1}{dt^2} = R_1\frac{d^2 i_2}{dt^2} - \frac{1}{C}\frac{di_1}{dt}$$

at $t=0+$

$$\frac{d^2 i_1}{dt^2}(0+) = \frac{d^2 i_2}{dt^2}(0+) - \frac{1}{RC}\frac{di_1}{dt}(0+)$$

$$= V\left[\frac{R_2}{L^2} - \frac{1}{R_1 LC}\right] - \frac{1}{R_1 C}\left[\frac{V}{L} - \frac{V}{R_1^2 C}\right]$$

$$= V\left[\frac{R_2}{L^2} - \frac{1}{R_1 LC} - \frac{1}{R_1 LC} + \frac{1}{R_1^3 C^2}\right]$$

$$\boxed{\frac{d^2 i_1}{dt^2}(0+) = V\left[\frac{R_2}{L^2} - \frac{2}{R_1 LC} + \frac{1}{R_1^3 C^2}\right]}$$